TBXT Hit Identification

End-to-end virtual screening pipeline

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Github: https://github.com/ndharms/tbxt-hackathon/tree/main

Strategy

Standard modern virtual screening: no single signal is strong enough on this target to rank candidates alone, so we combine several and filter at each stage.

Signals used:

- Zenodo SPR — ranker, ~0.65 AUROC ceiling
- Boltz — pose and site localization
- Vina — orthogonal physics
- Fragment substructure / similarity
- Physchem + PAINS + synthesis risk

How they're combined:

1. Zenodo ML model as a ranker, not a predictor
2. Boltz used primarily for pose, not raw ΔG
3. Vina for cross-check and anchor-residue contact
4. Physchem + PAINS + synthesis risk applied at every stage

Target and pocket plan

TBXT / Brachyury — T-box TF, key dependency in chordoma
DNA-binding domain (res 42–219), PDB 6F59
SGC TEP identifies 5 fragment pockets; we target **A, F, G**

TEP pocket	Anchor residues	Frags	Planned slots	QC survivors
A / A'	R180, V123, L91, I125, V173, I182	26	2	**70**
G	R54, E48, E50, L51, K76	10	1 -> 0	**0** (no pose QC)
F	Y88, D177, V173, I182	4	1 -> 2	**7**
~~D~~	G112, H100, P115	5	0 (bad Boltz pose)	—

Why these pockets (superposed all PDBs, scored contacts vs Newman signatures):

- **A (2 slots)** — most fragment evidence: 26 hits, the only site Newman progressed to µM SPR (thiazole → 8A7N, 14–20 µM). Boltz reproduces the crystal pose vs 5QS2.
- **G (1 slot)** — 10 hits, largest G177D hotspot. Polar pocket near C-term helix; angle adjacent to DNA interface, distinct from A.
- **F (1 slot, speculative)** — potential novel MoA: Y88 is the P300-interaction residue; pocket mediates KDM6 co-activator recruitment. Engages the G177D variant. Induced and buried, so we rely more on fragment similarity than Boltz scores.
- **D (dropped)** — Boltz pose tilted vs 5QS0 crystal, only 4–5 hits, DNA-competitive angle already covered by G.

Pipeline — end-to-end

```
onepot CORE                      SGC TEP fragments
       │                                 │
       ▼                                 │
[1] Downfilter: regex → RDKit            │
    substructure + physchem + PAINS      │
    + risk                               │
       │                                 ▼
       │        [1b] onepot API neighbor queries
       │             + physchem + PAINS + risk
       └──────────────────────────────────┘
                         │
                         ▼
             Combined screening library
                         ▼
        [2] XGBoost 12-booster ensemble → p_binder rank
                         ▼
        [3] Pocket-assign → split A / F / G (top 5000 each)
                         ▼
        [4] Diversity cluster (Morgan, Butina-style)
                         ▼
        [5] Boltz pose prediction + pose QC
              (site-localization rejects ~90–95%)
              QC survivors: **A: 70 · F: 7 · G: 0**
                         ▼
        [5b] Exclude Zenodo-similar compounds
              (ECFP4 Tanimoto > 0.85 to any SPR compound)
                         ▼
        [6] AutoDock Vina + anchor-residue check
              R180 (A) · Y88 (F) · E48/R54 (G — no survivors)
                         ▼
                Final 4 (2·A, 2·F — G slot reallocated)
```

Pocket G produced zero compounds that passed Boltz site-localization QC. Pocket A yielded 70 survivors; pocket F yielded 7. Slot plan updated: **2× A, 2× F**.

Modeling the Zenodo SPR data

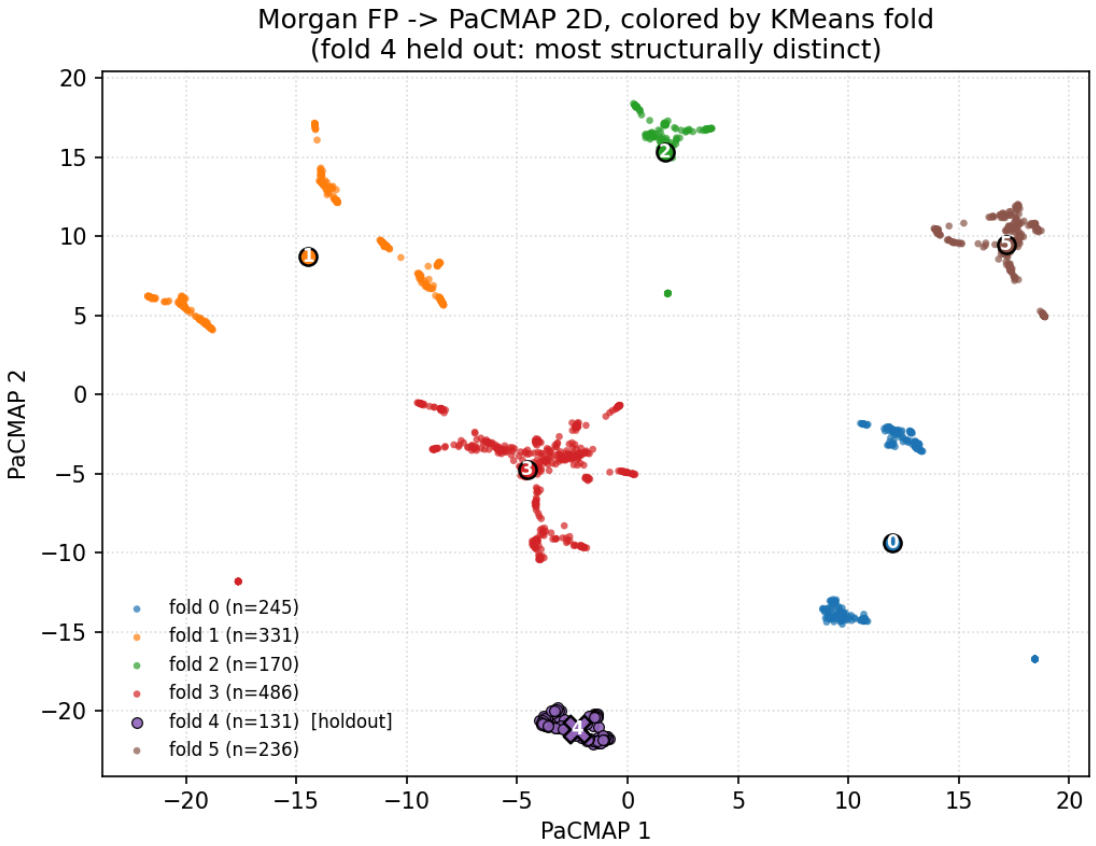
Data: 2,143 SPR records, 14 batches (Oct 2020 – Jan 2023), 1,545 unique compounds after cleaning.

The dataset is noisy.

- Batch date alone explains $R^2 \approx 0.15$ of pKD variance
- ~30% activity cliffs at Tanimoto ≥ 0.7; replicate std ≈ 0.92 pKD (~100× KD)
- Regression: all attempts had negative R^2 — a data ceiling, not a model ceiling
- Tried the **CheMeleon chemistry foundation model**; competitive but dropped for time

What we settled on: treat it as a binary ranker.

- **Excluded the mid-range KD gray zone** (pKD 3–5) — inside replicate noise
- **PaCMAP + KMeans6 folds** for structurally-distinct splits (Butina folds leak scaffolds)
- **12-booster XGBoost ensemble** (2 fingerprints × 6 leave-one-fold-out) on the filtered label



Training compounds in PaCMAP space, colored by KMeans6 fold. The ensemble trains 12 boosters leave-one-fold-out so every compound is scored by a booster that did not see it.

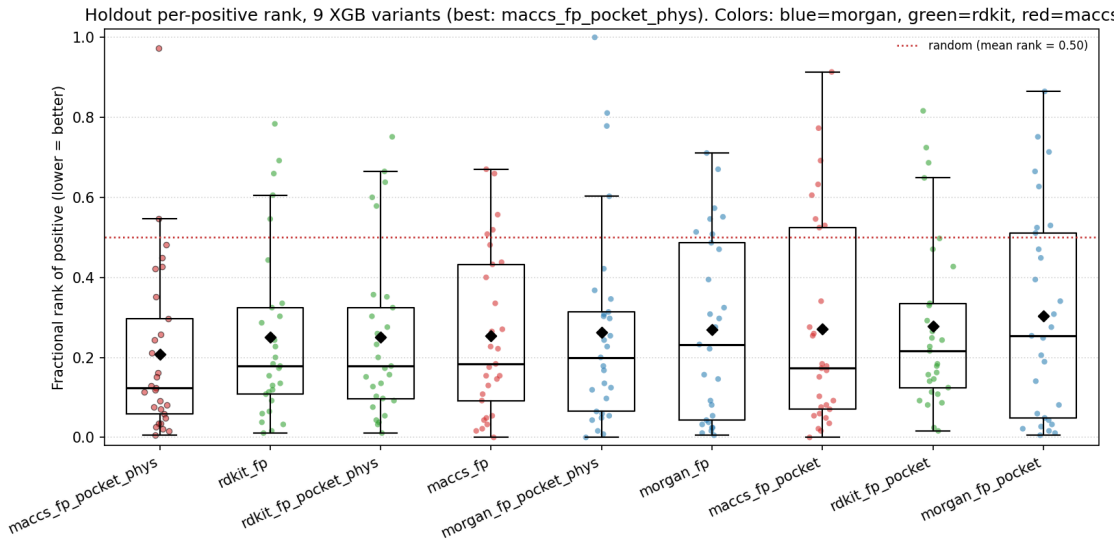
Model performance — positive ranking

For a ranker, the useful question is where the true positives land.

Evaluated on the try4 holdout (fold 3, n=186, 29 positives, 15.6% prevalence). For each positive, compute its fractional rank among the 186 holdout compounds after sorting by predicted score. Random = 0.5, perfect = 0.

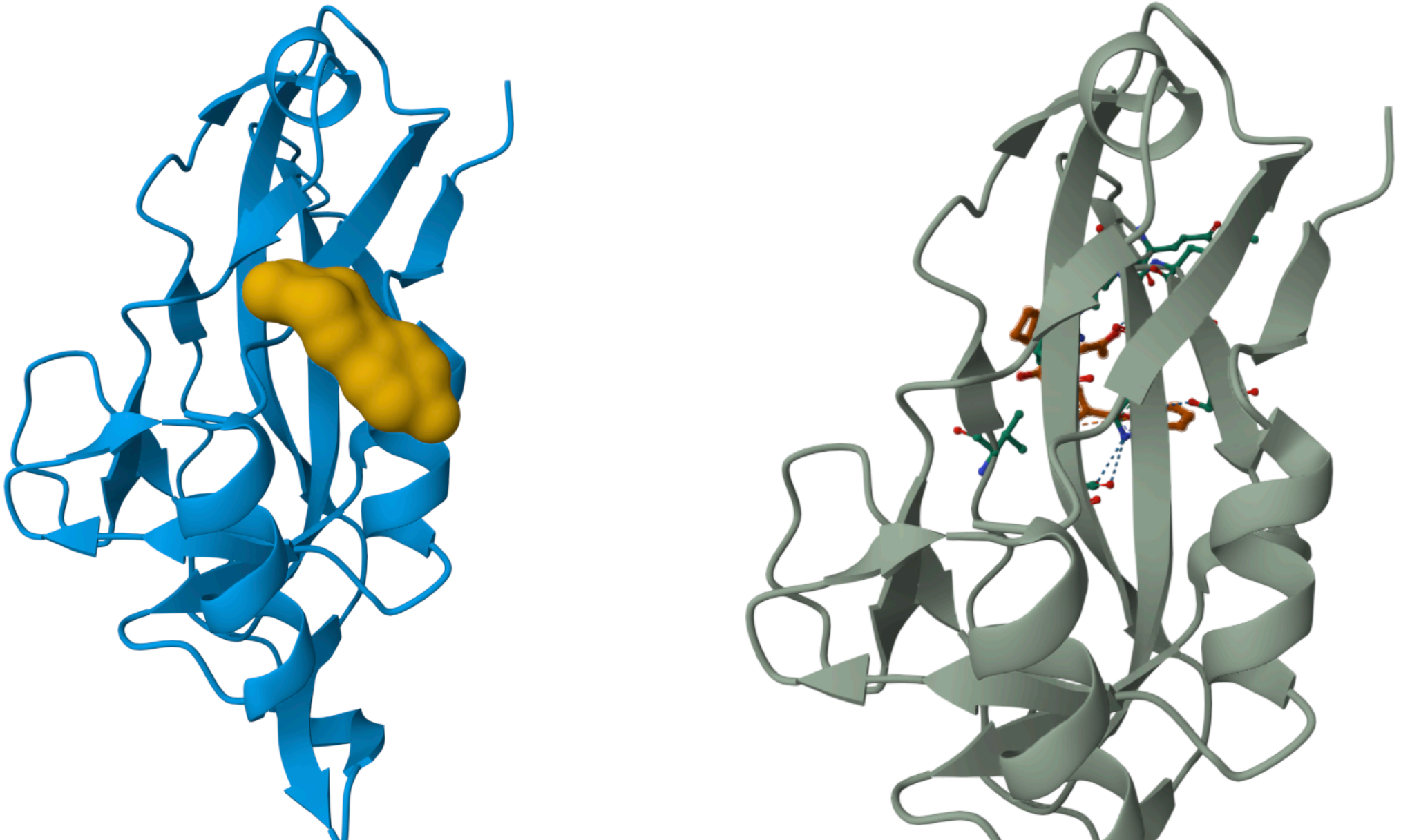
- `maccs_fp_pocket_phys` (the deployment recipe): median rank 0.124, mean 0.208 — positives land in roughly the top ~12% on average
- Tightest distribution among the 9 variants tested
- Non-fingerprint features contribute modestly (86% MACCS / 6% pocket / 8% physchem importance)

Morgan ranks a handful of positives very high but has a long right tail. MACCS + pocket + physchem is more evenly distributed. Including both in the ensemble covers both behaviors.



Per-positive fractional rank across 9 fingerprint × feature-set variants, holdout fold 3 (29 positives). Lower = better; random = 0.5 (grey line).

Boltz site-localization filter — in action



Final 4

Selected by **Vina rank order** within each pocket (2× A, 2× F; G reallocated after zero site-QC survivors). All four cleared the same pipeline — XGBoost rank → pocket assignment → Boltz site QC → Zenodo novelty → Vina + anchor-residue check.

Pocket	onepot ID	SMILES
A	`SM-AFB2BKPV`	`O=C(c1ccc2c(c1)CCCO2)N1CCC2(CC1)CC(=O)N(c1ccccc1)C2=O`
A	`SM-ZBGNCUEF`	`O=C(Cc1ccccc1)Nc1cccc(NC(=O)NCc2ccc3ncsc3c2)c1`
F	`SM-QZJTGDW5`	`Clc1cccc(Oc2nnc(Cc3ccccc3)c3ccccc23)c1`
F	`SM-68MRL1C4`	`O=C(Nc1cccc(OCc2ccccc2)c1)N1Cc2ccc(O)cc2C(O)C1`

Pocket labels follow the SGC TEP datasheet (pocket F = Newman D).

Appendices

Appendix A — Final 4 submission

Pocket	onepot ID	SMILES	Notes
A	`SM-AFB2BKPV`	`O=C(c1ccc2c(c1)CCCO2)N1CCC2(CC1)CC(=O)N(c1ccccc1)C2=O`	Amide donor for R180; extended hydrophobic vector for L91/V123/I125/V173.
A	`SM-ZBGNCUEF`	`O=C(Cc1ccccc1)Nc1cccc(NC(=O)NCc2ccc3ncsc3c2)c1`	Multiple donors/acceptors around the urea; benzothiazole fills the hydrophobic cluster.
F	`SM-QZJTGDW5`	`Clc1cccc(Oc2nnc(Cc3ccccc3)c3ccccc23)c1`	Compact aromatic fitting the induced Y88/D177 cavity; phthalazine N's near Y88.
F	`SM-68MRL1C4`	`O=C(Nc1cccc(OCc2ccccc2)c1)N1Cc2ccc(O)cc2C(O)C1`	Polar head for Y88/D177; benzyloxy tail extends into the buried cavity.

Common filter pass (all 4):

- XGBoost deployment ensemble rank (high p_binder)
- Fragment Tanimoto ≥ 0.35 in assigned pocket
- Boltz pose placed at the intended site (site-localization QC)
- ECFP4 Tanimoto ≤ 0.85 to Zenodo SPR set (novelty)
- AutoDock Vina anchor-residue contact (R180 for A; Y88 ± D177 for F)
- Chordoma physchem (LogP ≤ 6, HBD ≤ 6, HBA ≤ 12, MW ≤ 600)

Within-pocket ordering: Vina rank.
Pocket labels follow the SGC TEP datasheet (**F = Newman D**).

Appendix B — Stage 1 + 1b details (library build)

Downfilter 3.4B catalog

`scripts/downfilter_onepot/`

1. Coarse SMILES regex against canonical + Kekulized fragment SMILES + Bemis–Murcko scaffolds
2. RDKit worker pool: PAINS + bad-SMARTS + `HasSubstructMatch` vs 45 fragments
3. Hard filters: MW/LogP/HBD/HBA, 10–30 HAC, ≤ 5 rings, ≤ 2 fused, ≥ 1 fragment hit
4. Sharded parquet, rollup every 5M rows

Output: 897,431 compounds

onepot API nearest-neighbor queries

`scripts/query_onepot_neighbors.py`

- Per query pocket (A / F / G): 1 API call per SGC fragment
- Pocket A: 1000 neighbors/frag
- Pocket D/F/G: 2000 neighbors/frag
- ECFP4 (r=2, 2048) → route to best pocket
- Physchem + PAINS + synthesis-risk filters applied
- Fragment-substructure filter skipped (already similarity-selected)

Output: 51,560 unique compounds

Combined: `data/screening_library_combined.csv` **— 178,332 compounds**

Appendix C — Model metrics deep-dive

Deployment ensemble — 12 XGB boosters (2 FP × 6 leave-one-fold-out splits on PaCMAP-KMeans6).

Overall OOF (N=708, 12.3% prev):

- AUROC **0.764** · AUPRC **0.337** (random 0.123)
- Top-30 precision on try4 holdout: **17/30** = 57% (random ~16%)
- Mean fractional rank of positives: **0.208** (median 0.124)

Label choice: pKD > 5, filter 3–5 gray zone.
Try1 (no filter, top-quartile): holdout AUROC 0.48 (~random).
Try2 (filter gray zone): +0.06 to +0.12 OOF AUROC across variants.
Try4 (ablated FP × features): maccs+pocket+physchem was the best variant, and the only one where non-FP features meaningfully contribute (86% MACCS, 6% pocket, 8% physchem importance).

Why PaCMAP folds: Butina folds gave an inflated 97% AUROC / 89% precision from scaffold leakage. PaCMAP folds break scaffold-level correlation and give a more honest generalization estimate.

Feature pipeline (183-dim):

Slice	Source
167 cols	MACCS keys
8 cols	Pocket scores (Tanimoto + substruct, 4 pockets)
8 cols	Physchem (MW, LogP, HBD, HBA, HAC, rings, TPSA, rotB)

Ensemble probabilities used as **rank score only**, not calibrated P(binder).

Appendix C2 — Per-fold metrics

Morgan XGB booster family (per-fold OOF):

Held-out fold	N train	N test	Pos train	Pos test	OOF AUROC	OOF AUPRC
0	596	112	81	6	0.461	0.101
1	563	145	48	39	0.648	0.389
2	630	78	81	6	0.551	0.100
3	522	186	58	29	**0.800**	**0.410**
4	646	62	86	1	0.918*	0.167
5	583	125	81	6	0.637	0.282
Overall	—	708	—	87	**0.736**	**0.312**

*Fold 4 AUROC computed on 1 positive — uninterpretable in isolation. Fold 3 (29 positives) is the best-powered single-fold estimate.

Per-fold variance is large because positives are unevenly distributed across folds. Fold 0 (6 positives, AUROC 0.46) is a structurally distinct cluster the others don't sample well; the ensemble mean compensates. This is the cost of structure-aware splits on a small, batch-confounded dataset.

Appendix C3 — Pocket assignment details

Appendix D — Stage 4 + 5 (cluster + Boltz)

Top 5000 per pocket → diversity cluster

- Rank by `p_binder_ensemble_mean`
- Butina-style clustering on Morgan FP
- Redundant scaffolds removed *before* the expensive oracle runs

Model scoring is fast on 178k compounds. Boltz is the bottleneck, so we avoid spending it on scaffold twins.

Boltz pose prediction (Boltz Lab cloud)

Ran Boltz on the clustered top-5000-per-pocket. Pose QC is strict — the site-localization filter alone rejects **~90–95%** of Boltz outputs.

Post-QC survivors docked with Vina:

- **Pocket A: 10**
- **Pocket F: 7**
- **Pocket G: 0** (no compound placed at the intended site passed QC)

Pocket D not run (pose tilted vs crystal).

Site-localization filter: reject any compound whose Boltz pose places the ligand outside the intended pocket, allowing for induced-fit displacement. This is the dominant filter; it catches off-site placements that affinity score alone would not flag.

Per-site cautions carried into ranking:

Appendix E — Stage 6 + 7 (Vina + final selection)

AutoDock Vina · `docs/docking.md`

- Per-pocket receptor PDBs, box centered on reference fragment, 10 Å padding
- Multi-conformer, interaction analysis (H-bonds, π-stacking, hydrophobic, salt bridges)

Anchor-residue hard filter:

Pocket	Required contact
A	H-bond to **R180**
F	Contact to **Y88** (±D177)
~~G~~	**no QC survivors** — skipped

Fail the anchor check → drop regardless of Vina score.

Final 4 — each candidate must clear:

- ✅ XGBoost `p_binder_mean` ranks high
- ✅ Fragment Tanimoto ≥ 0.35 in assigned pocket
- ✅ Boltz pose matches the Newman co-crystal
- ✅ Vina score within 1 kcal/mol of per-pocket best
- ✅ Anchor contact present
- ✅ Chordoma physchem satisfied
- ✅ 4 distinct Bemis–Murcko scaffolds
- ✅ Max Tanimoto to Naar set < 0.6

Slot allocation (actual): 2× A · 2× F · 0× G
Planned 2·A + 1·G + 1·F; the G slot was reallocated to F after no G compounds passed Boltz site QC.

Appendix F — Caveats and references

Acknowledged limits

- SPR data has a ~0.65 AUROC ceiling for absolute prediction (batch effects, activity cliffs) — classifier used as a ranker
- Pocket F scores inflated by induced-fit; we rely on pose, not absolute ΔG
- O0P dual-binder means single-pocket routing is incomplete; multi-assignment fix documented
- CheMeleon foundation model tried and dropped for time
- Goal is fragment-informed, pose-sensible binders with multiple lines of agreement — not a potency breakthrough

Key numbers

Stage	Output
onepot CORE	3.4B
Downfilter survivors	897,431
API neighbors (unique)	51,560
Combined library	**?**
Top per pocket (pre-cluster)	5000
Boltz survivors (site-QC rejects ~90–95%)	A: 10 · F: 7 · G: 0
After Vina + anchor	final per-pocket shortlist
Submission	**4**

Combined library count needs verification — downfilter output + API-neighbor unique compounds don't sum cleanly to the CSV row count.

References: Newman et al. *Nat Commun* 2025 · SGC TBXT TEP · PDB 6F59 · onepot CORE

In-repo: `docs/HACKATHON_PLAN.md` · `docs/deployment-model.md` · `docs/docking.md` · `docs/brachyury-site-summary.md` · `docs/pocket-mapper-soundness.md` · `docs/sar-diagnostics-rjg/README.md`